

Replace `detector-id` with the detector ID of the member account for which you will delete the threat entity list, and other placeholder values that are *shown in red*.

To delete threat IP list

1. Run [DeleteThreatIntelSet](#). Make sure to provide the `detectorId` of the member account for which you want to delete this threat IP address list. To find the `detectorId` for your account and current Region, see the **Settings** page in the <https://console.aws.amazon.com/guardduty/> console, or run the [ListDetectors](#) API.
2. Alternatively, you can do this by running the following AWS Command Line Interface command and make sure to replace the `detector-id` with the detector ID of the member account for which you will delete the threat IP list.

```
aws guardduty delete-threat-intel-set \  
--detector-id 12abc34d567e8fa901bc2d34e56789f0 \  
--threat-intel-set-id d4b94fc952d6912b8f3060768example
```

Replace `detector-id` with the detector ID of the member account for which you will delete the threat entity list, and other placeholder values that are *shown in red*.

Sample 1 starts

Exporting generated GuardDuty findings to Amazon S3 buckets

GuardDuty retains the generated findings for a period of 90 days. GuardDuty exports the active findings to Amazon EventBridge (EventBridge). You can optionally export the generated findings to an Amazon Simple Storage Service (Amazon S3) bucket. This will help you to track the historical data of potentially suspicious activities in your account and evaluate whether the recommended remediation steps were successful.

Any new active findings that GuardDuty generates are automatically exported within about 5 minutes after the finding is generated. You can set the frequency for how often updates to the active findings are exported to EventBridge. The frequency that you select applies to the exporting of new occurrences of existing findings to EventBridge, your S3 bucket (when configured), and Detective (when integrated). For information about how GuardDuty aggregates multiple occurrences of existing findings, see [GuardDuty finding aggregation](#).

What is Amazon GuardDuty?

Amazon GuardDuty is a threat detection service that continuously monitors, analyzes, and processes AWS data sources and logs in your AWS environment. GuardDuty uses threat intelligence feeds, such as lists of malicious IP addresses and domains, file hashes, and machine learning (ML) models to identify suspicious and potentially malicious activity in your AWS environment. The following list provides an overview of potential threat scenarios that GuardDuty can help you detect:

- Compromised and exfiltrated AWS credentials.
- Data exfiltration and destruction that can lead to a ransomware event. Unusual patterns of login events in the supported engine versions of Amazon Aurora and Amazon RDS databases, that indicate anomalous behavior.
- Unauthorized cryptomining activity in your Amazon Elastic Compute Cloud (Amazon EC2) instances and container workloads.
- Presence of malware in your Amazon EC2 instances and container workloads, and newly uploaded files in your Amazon Simple Storage Service (Amazon S3) buckets.
- Operating system-level, networking, and file events that indicate unauthorized behavior on your Amazon Elastic Kubernetes Service (Amazon EKS) clusters, Amazon Elastic Container Service (Amazon ECS) - AWS Fargate tasks, and Amazon EC2 instances and container workloads.

The following video provides an overview of how GuardDuty helps you detect threats in your AWS environment.

[What is Amazon GuardDuty](#)

Contents

- [Features of GuardDuty](#)
- [PCI DSS Compliance](#)
- [Pricing in GuardDuty](#)
- [Accessing GuardDuty](#)

Features of GuardDuty

Here are some of the key ways in which Amazon GuardDuty can help you monitor, detect, and manage potential threats in your AWS environment.

Continuously monitors specific data sources and event logs

- **Foundational threat detection** – When you enable GuardDuty in an AWS account, GuardDuty automatically starts ingesting the foundational data sources associated with that account. These data sources include AWS CloudTrail management events, VPC flow logs (from Amazon EC2 instances), and DNS logs. You don't need to enable anything else for GuardDuty to start

analyzing and processing these data sources to generate associated security findings. For more information, see [GuardDuty foundational data sources](#).

- **Extended Threat Detection** – This capability detects multi-stage attacks that span foundational data sources, multiple types of AWS resources, and time, within an AWS account. There might be multiple events in your account that, individually, don't present themselves as a clear threat. However, when these events are observed in a sequence that is indicative of a suspicious activity, GuardDuty identifies it as an attack sequence. GuardDuty notifies you by generating the associated attack sequence finding type to provide details about the observed attack sequence.

With no additional cost associated with it, Extended Threat Detection is automatically enabled for each AWS account when they enable GuardDuty. This capability doesn't require you to enable any use-case focused protection plan. However, to increase the breadth of security to your Amazon S3 resources, GuardDuty recommends enabling S3 Protection in your account. This will help Extended Threat Detection to identify multi-stage attacks that potentially impact your Amazon S3 resources.

For more information about how this capability works and what threat scenarios it covers, see [GuardDuty Extended Threat Detection](#).

- **Investigation (Preview)** – GuardDuty Investigation provides AI-powered security analysis of your GuardDuty findings. You can create investigations to analyze specific findings, accounts, or your entire organization, and receive structured investigation summaries with risk assessment, confidence scoring, MITRE ATT&CK® classification, and actionable next steps. For more information, see [GuardDuty Investigation \(Preview\)](#).
- **Use-case focused GuardDuty protection plans** – For enhanced threat detection visibility into the security of your AWS environment, GuardDuty offers dedicated protection plans that you can choose to enable. Protection plans help you monitor logs and events from other AWS services. These sources include EKS audit logs, RDS login activity, Amazon S3 data events in CloudTrail, EBS volumes, Runtime Monitoring across Amazon EKS, Amazon EC2, and Amazon ECS-Fargate, Lambda network activity logs, and AWS CloudTrail data events from Amazon Bedrock, Amazon Bedrock AgentCore, and Amazon SageMaker AI for AI Protection. GuardDuty consolidates these log and event sources under the term - [Features](#). You can enable one or more dedicated protection plans in a supported AWS Region at any time. GuardDuty will start monitoring, processing, and analyzing the activities based on which protection plan you enable. For more information about each protection plan and how it works, see the corresponding protection plan document.

When you enable GuardDuty for the first time, GuardDuty automatically enables all protection plans except Runtime Monitoring for your account, and includes them in the 30-day free trial. You can disable any protection plan at any time. If you are an existing GuardDuty customer, a protection plan that launches after you enabled GuardDuty is not enabled automatically, and you can choose to enable it. For information about which protection plans are enabled by default, see [Using GuardDuty 30-day free trial](#).

Protection plan	Description
AI Protection	Detects threats to your AI workloads that use Amazon Bedrock, Amazon Bedrock AgentCore, and Amazon SageMaker AI, such as anomalous model invocations and cost harvesting attacks.
S3 Protection	Identifies potential security risks such as data exfiltration and destruction attempts in your Amazon S3 buckets.
EKS Protection	EKS Audit Log Monitoring analyzes Kubernetes audit logs from your Amazon EKS clusters for potentially suspicious and malicious activities.
Runtime Monitoring	Monitors and analyzes operating system-level events on your Amazon EKS, Amazon EC2, and Amazon ECS (including AWS Fargate), to detect potential runtime threats.
Malware Protection for EC2	Detects potential presence of malware by scanning the Amazon EBS volumes associated with your Amazon EC2 instances. There is an option to use this feature on-demand.
Malware Protection for S3	Detects potential presence of malware in the newly uploaded objects within your Amazon S3 buckets. There is an option to use this feature on-demand.

Protection plan	Description
Malware Protection for AWS Backup	Detects potential presence of malware by scanning backup resources, including EBS Snapshots, EC2 AMI's, and AWS Backup recovery points. There is an option to use this feature on-demand.
RDS Protection	Analyzes and profiles your RDS login activity for potential access threats to the supported Amazon Aurora and Amazon RDS databases.
Lambda Protection	Monitors Lambda network activity logs, starting with VPC flow logs, to detect threats to your AWS Lambda functions . Examples of these potential threats include cryptomining and communicating with malicious servers.

Enable Malware Protection for S3 independently

GuardDuty offers flexibility to use Malware Protection for S3 independently, without enabling the Amazon GuardDuty service. For more information about getting started with only Malware Protection for S3, see [GuardDuty Malware Protection for S3](#). To use all other protection plans, you must enable the GuardDuty service.

Manage multiple-account environment

You can manage a multiple-account AWS environment by using either AWS Organizations (recommended) or legacy invitation method. For more information, see [Multiple accounts in GuardDuty](#).

Generates security findings for detected threats

When GuardDuty detects potential security threats associated with your AWS resources, it starts generating security findings that provide information about the potentially compromised resource. After you enable GuardDuty in your account, generate [Sample findings](#) to view the associated [Finding details](#). For a complete list of security findings, see [GuardDuty finding types](#).

With GuardDuty, you can also use a tester script that generates specific GuardDuty security findings to understand how to review and respond to GuardDuty findings. For more information, see [Test GuardDuty findings in dedicated accounts](#).

Assessing and managing security findings

GuardDuty consolidates your security findings across accounts and displays results in the Summary dashboard on the GuardDuty console. You can also retrieve findings through the AWS Security Hub CSPM API, AWS Command Line Interface, or AWS SDK. With a holistic view of your current security status, you can identify trends and potential issues, and take necessary remediation steps. For more information, see [Managing GuardDuty findings](#).

Integrate with related AWS security services

To further help you analyze and investigate the security trends in your AWS environment, consider using the following AWS security-related services in combination with GuardDuty.

- **AWS Security Hub CSPM** – This service gives you a comprehensive view of the security state of your AWS resources and helps you check your AWS environment against security industry standards and best practices. It does this partly by consuming, aggregating, organizing, and prioritizing your security findings from multiple AWS services (including Amazon Macie) and supported AWS Partner Network (APN) products. Security Hub CSPM helps you analyze your security trends and identify the highest priority security issues across your AWS environment.

For information about using GuardDuty and Security Hub CSPM together, see [Integrating GuardDuty with AWS Security Hub CSPM](#). To learn more about Security Hub CSPM, see the [AWS Security Hub User Guide](#).

- **Amazon Detective** – This service helps you analyze, investigate, and quickly identify the root cause of security findings or suspicious activities. Detective automatically collects log data from your AWS resources. It then uses machine learning, statistical analysis, and graph theory to generate visualizations that help you to conduct faster and more efficient security investigations. The Detective prebuilt data aggregations, summaries, and context help you analyze and determine the nature and extent of potential security issues.

For information about using GuardDuty and Detective together, see [Integrating GuardDuty with Amazon Detective](#). To learn more about Detective, see the [Amazon Detective User Guide](#).

- **Amazon EventBridge** – This service helps you receive notifications and respond to GuardDuty security findings in near-real time. GuardDuty creates an event when there is a change in the findings. You can choose how frequently you want to receive the notifications from EventBridge. For more information, see [What is Amazon EventBridge](#) in the *Amazon EventBridge User Guide*.

PCI DSS Compliance

GuardDuty supports the processing, storage, and transmission of credit card data by a merchant or service provider, and has been validated as being compliant with Payment Card Industry (PCI) Data Security Standard (DSS). For more information about PCI DSS, including how to request a copy of the AWS PCI Compliance Package, see [PCI DSS Level 1](#).

For more information, see [New third-party test compares Amazon GuardDuty to network intrusion detection systems](#) in the *AWS Security Blog*.

Pricing in GuardDuty

This section focuses on the AWS Free Tier model that GuardDuty uses for various protection plans, and how you can view estimated and actual usage costs. If you are looking for the pricing details associated with all the protection plans across supported Regions, see [GuardDuty pricing](#).

AWS Free Tier

AWS Free Tier helps you explore and try out AWS services free of charge up to specified limits for each service. There are three categories – 12 months free, always free, and short-term free trials. Amazon GuardDuty belongs to the short-term free trial category and offers a 30-day free trial. When you continue using GuardDuty after this free trial ends, you start incurring cost based on how you use this service.

¹Exception to GuardDuty 30-day free trial

On-demand malware scan (under Malware Protection for EC2) and Malware Protection for S3 don't fall into the GuardDuty 30-day short term free trial category. Malware Protection for S3 includes a Free Tier whereas the On-demand malware scan follows a pay-as-you-use cost model. There is no 30-day free trial or a Free Tier cost model with On-demand malware scan.

Using GuardDuty 30-day free trial

When using GuardDuty for the first time in an AWS Region, your AWS account is automatically enrolled in a 30-day free trial in that Region. Some of the protection plans will also get enabled automatically and are included in the 30-day free trial. Because GuardDuty is a regional service, when you enable it for the first time in a different Region, your account will get a 30-day free trial of GuardDuty in that Region. When working with multiple accounts in a GuardDuty organization, each account gets its own 30-day free trial.

Use the following table to review which protection plans are enabled by default with GuardDuty, and their free trial availability.

Protection plan	Enabled by default with GuardDuty	Separate free trial availability ²	
EKS Protection	Yes	Yes	
S3 Protection	Yes	Yes	
Runtime Monitoring	No	Yes	
Malware Protection for EC2 – GuardDuty-initiated malware scan	Yes	Yes	
Malware Protection for EC2 – On-demand malware scan in GuardDuty	No	No ¹	
GuardDuty Malware Protection for S3	No	No ¹	
RDS Protection	Yes	Yes	
Lambda Protection	Yes	Yes	
Malware Protection for AWS Backup	No	No	

²When you enable GuardDuty for the first time, protection plans (except Runtime Monitoring) are automatically enabled and included in the initial 30-day free trial. When an existing GuardDuty account enables a new protection plan after their initial GuardDuty free trial has expired, then that protection plan comes with its own 30-day free trial. For more information about free trials for protection plans, see the document associated with each protection plan.

View estimated usage cost during free trial – During 30-day free trial of GuardDuty and potentially a protection plan, GuardDuty provides estimated usage cost for your account. If you're a delegated GuardDuty administrator account, you can view the total estimated usage cost and account-level breakdown for all the member accounts that have enabled GuardDuty. For more information, see [Monitoring GuardDuty Usage and Estimating Costs](#).

Usage cost after free trial ends – When you continue using GuardDuty or any of its protection plans after the free trial ends, you will start incurring associated usage costs. To view your bill, navigate to **Cost Explorer** in the <https://console.aws.amazon.com/costmanagement/> console. For more information about AWS account billing, see the [AWS Billing User Guide](#).

Using Malware Protection for S3 with Free Tier

Malware Protection for S3 includes a Free Tier associated with your AWS accounts. For more information, see [Pricing and usage cost for Malware Protection for S3](#).

Accessing GuardDuty

Amazon GuardDuty is available in most AWS Regions. For a list of Regions where GuardDuty is currently available, see [Regions and endpoints](#).

You can use GuardDuty in any of the following ways:

GuardDuty console

<https://console.aws.amazon.com/guardduty/>

The console is a browser-based interface to access and use GuardDuty. The GuardDuty console provides access to your GuardDuty account, data, and resources.

AWS Command Line Interface

With AWS Command Line Interface (AWS CLI), you can issue commands at your system's command line to perform GuardDuty tasks and AWS tasks. The AWS CLI commands are useful if you want to build scripts that perform tasks.

For information about installing and using AWS CLI, see [AWS Command Line Interface User Guide](#). To view the available AWS CLI commands for GuardDuty, see [AWS CLI Command Reference](#).

GuardDuty HTTPS API

You can access GuardDuty and AWS programmatically by using the GuardDuty HTTPS API, which lets you issue HTTPS requests directly to the service. For more information, see the [Amazon GuardDuty API Reference](#).

AWS SDKs

AWS provides software development kits (SDKs) that consist of libraries and sample code for various programming languages and platforms (Java, Python, Ruby, .NET, iOS, Android, and more). The SDKs provide a convenient way to create programmatic access to GuardDuty. For information about the AWS SDKs, including how to download and install them, see [Tools for Amazon Web Services](#).

When you configure settings to export findings to an Amazon S3 bucket, GuardDuty uses AWS Key Management Service (AWS KMS) to encrypt the findings data in your S3 bucket. This requires you to add permissions to your S3 bucket and the AWS KMS key so that GuardDuty can use them to export findings in your account.

Contents

- [Considerations](#)
- [Step 1 – Permissions required to export findings](#)
- [Step 2 – Attaching policy to your KMS key](#)
- [Step 3 – Attaching policy to Amazon S3 bucket](#)
- [Step 4 - Exporting findings to an S3 bucket \(Console\)](#)
- [Step 5 – Setting frequency to export updated active findings](#)

Considerations

Before proceeding with the prerequisites and steps to export findings, consider the following key concepts:

- **Export settings are regional** – You need to configure export options in each Region where you use GuardDuty.
- **Exporting findings to Amazon S3 buckets in different AWS Regions (cross-Region)** – GuardDuty supports the following export settings:
 - Your Amazon S3 bucket or object, and AWS KMS key must belong to the same AWS Region.
 - For the findings generated in a commercial Region, you can choose to export these findings to an S3 bucket in any commercial Region. However, you can't export these findings to an S3 bucket in an opt-in Region.
 - For the findings generated in an opt-in Region, you can choose to export these findings to the same opt-in Region where they're generated or any commercial Region. However, you can't export findings from one opt-in Region to another opt-in Region.
- **Permissions to export findings** – To configure settings for exporting active findings, your S3 bucket must have permissions that allows GuardDuty to upload objects. You must also have an AWS KMS key that GuardDuty can use to encrypt findings.
- **Archived findings are not exported** – The default behavior is that the archived findings, including new instances of suppressed findings, are not exported.

When a GuardDuty finding gets generated as *Archived*, you will need to *Unarchive* it. This changes the **Filter finding status** to **Active**. GuardDuty exports the updates to the existing unarchived findings based on how you configure [Step 5 – Frequency for exporting findings](#).

- **GuardDuty administrator account can export findings generated in associated member accounts** – When you configure export findings in an administrator account, all the findings from the associated member accounts that are generated in the same Region are also exported to the same location that you configured for the administrator account. For more information, see [Understanding the relationship between GuardDuty administrator account and member accounts](#).

Step 1 – Permissions required to export findings

When you configure settings for exporting findings, you select an Amazon S3 bucket where you can store the findings and an AWS KMS key to use for data encryption. In addition to permissions for GuardDuty actions, you must also have permissions to the following actions to successfully configure settings to export findings:

- `s3:GetBucketLocation`
- `s3:PutObject`

If you need to export the findings to a specific prefix in your Amazon S3 bucket, you must also add the following permissions to the IAM role:

- `s3:GetObject`
- `s3:ListBucket`

Step 2 – Attaching policy to your KMS key

GuardDuty encrypts the findings data in your bucket by using AWS Key Management Service. To successfully configure the settings, you must first give GuardDuty permission to use a KMS key. You can grant the permissions by [attaching the policy](#) to your KMS key.

When you use a KMS key from another account, you need to apply the key policy by logging in to the AWS account that owns the key. When you configure the settings to export findings, you'll also need the key ARN from the account that owns the key.

To modify the KMS key policy for GuardDuty to encrypt your exported findings

1. Open the AWS KMS console at <https://console.aws.amazon.com/kms>.
2. To change the AWS Region, use the Region selector in the upper-right corner of the page.
3. Select an existing KMS key or perform the steps to [Create a new key](#) in the *AWS Key Management Service Developer Guide*, that you will use to encrypt the exported findings.

Note

The AWS Region of your KMS key and the Amazon S3 bucket must be the same.

You can use the same S3 bucket and KMS key pair to export the findings from any applicable Region. For more information, see [Considerations](#) for exporting findings across Regions.

4. In the **Key policy** section, choose **Edit**.

If **Switch to policy view** is displayed, choose it to display the **Key policy**, and then choose **Edit**.

5. Copy the following policy block to your KMS key policy, to grant GuardDuty permission to use your key.

```
{
  "Sid": "AllowGuardDutyKey",
  "Effect": "Allow",
  "Principal": {
    "Service": "guardduty.amazonaws.com"
  },
  "Action": "kms:GenerateDataKey",
  "Resource": "KMS key ARN",
  "Condition": {
    "StringEquals": {
      "aws:SourceAccount": "123456789012",
      "aws:SourceArn":
        "arn:aws:guardduty:Region2:123456789012:detector/SourceDetectorID"
    }
  }
}
```

6. Edit the policy by replacing the following values that are formatted in *red* in the policy example:

1. Replace *KMS key ARN* with the Amazon Resource Name (ARN) of the KMS key. To locate the key ARN, see [Finding the key ID and ARN](#) in the *AWS Key Management Service Developer Guide*.
2. Replace *123456789012* with the AWS account ID that owns the GuardDuty account exporting the findings.
3. Replace *Region2* with the AWS Region where the GuardDuty findings are generated.
4. Replace *SourceDetectorID* with the detectorID of the GuardDuty account in the specific Region where the findings generated.

To find the detectorId for your account and current Region, see the **Settings** page in the <https://console.aws.amazon.com/guardduty/> console, or run the [ListDetectors](#) API.

Note

If you're using GuardDuty in an opt-in Region, replace the value for the "Service" with the Regional endpoint for that Region. For example, if you're using GuardDuty in the Middle East (Bahrain) (me-south-1) Region, replace "Service": "guardduty.amazonaws.com" with "Service": "guardduty.me-south-1.amazonaws.com". For information about endpoints for each opt-in Region, see [GuardDuty endpoints and quotas](#).

7. If you added the policy statement before the final statement, add a comma before adding this statement. Make sure that the JSON syntax of your KMS key policy is valid.

Choose **Save**.

8. (Optional) copy the key ARN to a notepad for use in the later steps.

Step 3 – Attaching policy to Amazon S3 bucket

Add permissions to the Amazon S3 bucket to which you will export findings so that GuardDuty can upload objects to this S3 bucket. Independent of using an Amazon S3 bucket that belongs to either your account or in a different AWS account, you must add these permissions.

If at any point in time, you decide to export findings to a different S3 bucket, then to continue exporting findings, you must add permissions to that S3 bucket and configure the export findings settings again.

If you do not already have an Amazon S3 bucket where you want to export these findings, see [Creating a bucket](#) in the *Amazon S3 User Guide*.

To attach permissions to your S3 bucket policy

1. Perform the steps under [To create or edit a bucket policy](#) in the *Amazon S3 User Guide*, until the **Edit bucket policy** page appears.
2. The **example policy** shows how grant GuardDuty permission to export findings to your Amazon S3 bucket. If you change the path after you configure export findings, then you must modify the policy to grant permission to the new location.

Copy the following **example policy** and paste it into the **Bucket policy editor**.

If you added the policy statement before the final statement, add a comma before adding this statement. Make sure that the JSON syntax of your KMS key policy is valid.

S3 bucket example policy

JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "Allow GetBucketLocation",
      "Effect": "Allow",
      "Principal": {
        "Service": "guardduty.amazonaws.com"
      },
      "Action": "s3:GetBucketLocation",
      "Resource": "arn:aws:s3:::amzn-s3-demo-bucket",
      "Condition": {
        "StringEquals": {
          "aws:SourceAccount": "123456789012",
          "aws:SourceArn": "arn:aws:guardduty:us-east-2:123456789012:detector/SourceDetectorID"
        }
      }
    }
  ],
  {
```

```

        "Sid": "Allow PutObject",
        "Effect": "Allow",
        "Principal": {
            "Service": "guardduty.amazonaws.com"
        },
        "Action": "s3:PutObject",
        "Resource": "arn:aws:s3:::amzn-s3-demo-bucket[optional prefix]/
    *",
        "Condition": {
            "StringEquals": {
                "aws:SourceAccount": "123456789012",
                "aws:SourceArn": "arn:aws:guardduty:us-
east-2:123456789012:detector/SourceDetectorID"
            }
        }
    },
    {
        "Sid": "Deny unencrypted object uploads",
        "Effect": "Deny",
        "Principal": {
            "Service": "guardduty.amazonaws.com"
        },
        "Action": "s3:PutObject",
        "Resource": "arn:aws:s3:::amzn-s3-demo-bucket[optional prefix]/
    *",
        "Condition": {
            "StringNotEquals": {
                "s3:x-amz-server-side-encryption": "aws:kms"
            }
        }
    },
    {
        "Sid": "Deny incorrect encryption header",
        "Effect": "Deny",
        "Principal": {
            "Service": "guardduty.amazonaws.com"
        },
        "Action": "s3:PutObject",
        "Resource": "arn:aws:s3:::amzn-s3-demo-bucket[optional prefix]/
    *",
        "Condition": {
            "StringNotEquals": {

```

```

        "s3:x-amz-server-side-encryption-aws-kms-key-id":
        "arn:aws:kms:us-east-2:111122223333:key/a1b2c3d4-5678-90ab-cdef-
        EXAMPLE11111"
    },
    {
        "Sid": "Deny non-HTTPS access",
        "Effect": "Deny",
        "Principal": "*",
        "Action": "s3:*",
        "Resource": "arn:aws:s3:::amzn-s3-demo-bucket[optional prefix]/
        *",
        "Condition": {
            "Bool": {
                "aws:SecureTransport": "false"
            }
        }
    }
]
}

```

3. Edit the policy by replacing the following values that are formatted in *red* in the policy example:

1. Replace *Amazon S3 bucket ARN* with the Amazon Resource Name (ARN) of the Amazon S3 bucket. You can find the **Bucket ARN** on the **Edit bucket policy** page in the <https://console.aws.amazon.com/s3/> console.
2. Replace *123456789012* with the AWS account ID that owns the GuardDuty account exporting the findings.
3. Replace *us-east-2* with the AWS Region where the GuardDuty findings are generated.
4. Replace *SourceDetectorID* with the detectorID of the GuardDuty account in the specific Region where the findings generated.

To find the detectorId for your account and current Region, see the **Settings** page in the <https://console.aws.amazon.com/guardduty/> console, or run the [ListDetectors](#) API.

5. Replace *[optional prefix]* part of the *S3 bucket ARN/[optional prefix]* placeholder value with an optional folder location to which you want to export the findings. For more information about the use of prefixes, see [Organizing objects using prefixes](#) in the *Amazon S3 User Guide*.

When you provide an optional folder location that doesn't exist already, GuardDuty will create that location only if the account associated with the S3 bucket is the same as the account exporting the findings. When you export findings to an S3 bucket that belongs to another account, the folder location must exist already.

6. Replace *KMS key ARN* with the Amazon Resource Name (ARN) of the KMS key associated with the encryption of the findings exported to the S3 bucket. To locate the key ARN, see [Finding the key ID and ARN](#) in the *AWS Key Management Service Developer Guide*.

Note

If you're using GuardDuty in an opt-in Region, replace the value for the "Service" with the Regional endpoint for that Region. For example, if you're using GuardDuty in the Middle East (Bahrain) (me-south-1) Region, replace "Service": "guardduty.amazonaws.com" with "Service": "guardduty.me-south-1.amazonaws.com". For information about endpoints for each opt-in Region, see [GuardDuty endpoints and quotas](#).

4. Choose **Save**.

Step 4 - Exporting findings to an S3 bucket (Console)

GuardDuty permits you to export findings to an existing bucket in another AWS account.

When creating a new S3 bucket or choosing an existing bucket in your account, you can add an optional prefix. When configuring export findings, GuardDuty creates a new folder in the S3 bucket for your findings. The prefix will be appended to the default folder structure that GuardDuty created. For example, the format of the optional prefix `/AWSLogs/123456789012/GuardDuty/Region`.

The entire path of the S3 object will be `amzn-s3-demo-bucket/prefix-name/UUID.jsonl.gz`. The UUID is randomly generated and doesn't represent the detector ID or the finding ID.

Important

The KMS key and S3 bucket must be in the same Region.

Before completing these steps, make sure you have attached the respective policies to your KMS key and existing S3 bucket.

To configure export findings

1. Open the GuardDuty console at <https://console.aws.amazon.com/guardduty/>.
2. In the navigation pane, choose **Settings**.
3. On the **Settings** page, under **Findings export options**, for **S3 bucket**, choose **Configure now** (or **Edit**, as needed).
4. For **S3 bucket ARN**, enter the **bucket ARN**. To find the bucket ARN, see [Viewing the properties for an S3 bucket](#) in the *Amazon S3 User Guide*.
5. For **KMS key ARN**, enter the **key ARN**. To locate the key ARN, see [Finding the key ID and ARN](#) in the *AWS Key Management Service Developer Guide*.
6. **Attach policies**
 - Perform the steps to attach the S3 bucket policy. For more information, see [Step 3 – Attaching policy to Amazon S3 bucket](#).
 - Perform the steps to attach the KMS key policy. For more information, see [Step 2 – Attaching policy to your KMS key](#).
7. Choose **Save**.

Step 5 – Setting frequency to export updated active findings

Configure the frequency for exporting updated active findings as appropriate for your environment. By default, updated findings are exported every 6 hours. This means that any findings that are updated after the most recent export are included in the next export. If updated findings are exported every 6 hours and the export occurs at 12:00, any finding that you update after 12:00 is exported at 18:00.

To set the frequency

1. Open the GuardDuty console at <https://console.aws.amazon.com/guardduty/>.
2. Choose **Settings**.

3. In the **Findings export options** section, choose **Frequency for updated findings**. This sets the frequency for exporting updated Active findings to both EventBridge and Amazon S3. You can choose from the following:
 - **Update EventBridge and S3 every 15 minutes**
 - **Update EventBridge and S3 every 1 hour**
 - **Update EventBridge and S3 every 6 hours (default)**
4. Choose **Save changes**.

Sample 1 ends

Processing GuardDuty findings with Amazon EventBridge

GuardDuty automatically publishes (sends) findings as events to Amazon EventBridge (formerly Amazon CloudWatch Events), a serverless event bus service. EventBridge delivers a stream of near real-time data from applications and services to targets such as Amazon Simple Notification Service (Amazon SNS) topics, AWS Lambda functions, and Amazon Kinesis streams. For more information, see [Amazon EventBridge User Guide](#).

EventBridge enables automated monitoring and processing of GuardDuty findings by receiving [events](#). EventBridge receives events for both newly generated findings and aggregated findings, where subsequent occurrences of an existing finding are combined with the original. Every GuardDuty finding is assigned a finding ID, and GuardDuty creates an EventBridge event for every finding with a unique finding ID. For information on how aggregation works in GuardDuty, see [GuardDuty finding aggregation](#).

In addition to automated monitoring and processing, use of EventBridge enables longer-term retention of your findings data. GuardDuty stores findings for 90 days. With EventBridge, you can send findings data to your preferred storage platform and store the data for as long as you like. To retain findings for a longer duration, GuardDuty supports [Exporting generated findings to Amazon S3](#).

Topics

- [Understanding EventBridge notification frequency in GuardDuty](#)
- [Set up an Amazon SNS topic and endpoint \(Email, Slack, and Amazon Chime\)](#)
- [Using Amazon EventBridge for GuardDuty findings](#)
- [Creating an EventBridge rule for GuardDuty findings](#)
- [EventBridge rule for GuardDuty multi-account environments](#)